

CMPS 2200 Assignment 02

Answers

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Place all written answers from `assignment-02.md` here for easier grading.

1. Asymptotic notation

a) $T(n) = 2T(n/3) + 1$

- Work at level 0: 1
- Work at level 1: 2
- Work at level 2: 4

Therefore, $T(n) = \sum_{i=0}^{\log_3 n} 2^i = 2^{\log_3 n+1} - 1 = 2n^{\log_3 2} - 1 \in O(n^{\log_3 2}) \in O(n)$

b) $T(n) = 5T(n/4) + n$

- Work at level 0: n
- Work at level 1: $\frac{5}{4}n$

Since $\frac{5}{4} > 1$, we know that this recurrence is leaf-dominated.

The number of leaves: $5^{\log_4 n} = n^{\log_4 5}$

Therefore, $T(n) = n^{\log_4 5} \approx n^{1.16} \in O(n^{1.16})$

c) $T(n) = 7T(n/7) + n$

- Work at level 0: n
- Work at level 1: $\frac{7}{7}n = n$

Since the amount of work at every level is the same, this recurrence is balanced.

Therefore, $T(n) = n \log_7 n \in O(n \log n)$

d) $T(n) = 9T(n/3) + n^2$

- Work at level 0: n^2
- Work at level 1: $9 \frac{n^2}{9} = n^2$
- $T(2) = 9^2 \frac{n^2}{9^2} = n^2$

Since the amount of work at every level is the same, this recurrence is balanced.

Therefore, $T(n) = n^2 \log_3 n \in O(n^2 \log n)$

e) $T(n) = 8T(n/2) + n^3$

- Work at level 0: n^3
- Work at level 1: $8 \frac{n^3}{8} = n^3$

Since the amount of work at every level is the same, this recurrence is balanced.

Therefore, $T(n) = n^3 \log_2 n \in O(n^3 \log n)$

f) $T(n) = 49T(n/25) + n^{3/2} \log n$

- Work at level 0: $n^{3/2} \log n$

- Work at level 1: $49 \frac{n^{3/2}}{25^{3/2}} \log \frac{n}{25}$

Since $\frac{49}{125} < 1$ and $\frac{n}{25} < n$, we know that this recurrence is root-dominated.

Therefore, $T(n) = n^{\frac{3}{2}} \log n \in O(n^{\frac{3}{2}} \log n)$

g) $T(n) = T(n-1) + 2$

- Work at level 0: 2
- Work at level 1: 2

Since the amount of work at every level is the same, this recurrence is balanced.

Therefore, $T(n) = 2n \in O(n)$

h) $T(n) = T(n-1) + n^c$, with $c \geq 1$

Since $c \geq 1, \forall i < n, i^c < n^c$

$$T(n) = \sum_{i=0}^n i^c < n \cdot n^c = n^{c+1}$$

Therefore, $T(n) \in O(n^{c+1})$

i) $T(n) = T(\sqrt{n}) + 1$

- Work at level 0: 1
- Work at level 1: 1
- Work at level 2: 1

Since the amount of work at every level is the same, this recurrence is balanced.

And the height of the tree is $\log \log n$

Therefore, $T(n) = O(\log \log n)$

2. Algorithms Comparison

- Algorithm A: $T(n) = 5T(n/2) + n$

- Work at level 0: n
- Work at level 1: $\frac{5}{2}n$

Since $\frac{5}{2} > 1$, this recurrence is leaf-dominated.

The number of leaves: $5^{\log_2 n} = n^{\log_2 5} \approx n^{2.32}$

$$\Rightarrow T(n) = n^{\log_2 5} \in O(n^{2.32})$$

- Algorithm B: $T(n) = 2T(n-1) + 1$

$$\begin{aligned} - T(n) &= 2T(n-1) + 1 = 2(2T(n-2) + 1) + 1 = 4T(n-2) + 2 + 1 = 4(2T(n-3) + 1) + 2 + 1 = \\ &8T(n-3) + 4 + 2 + 1 = \dots = 2^{n-1}T(1) + \dots + 4 + 2 + 1 = \sum_{i=0}^{n-1} 2^i = 2^n - 1 = O(2^n) \end{aligned}$$

- Algorithm C: $T(n) = 9T(n/3) + n^2$

- Work at level 0: n^2
- Work at level 1: $9 \frac{n^2}{9} = n^2$

Since the amount of work at every level is the same, this recurrence is balanced.

Therefore, $T(n) = n^2 \log_3 n \in O(n^2 \log n)$

- **Conclusion:** Choose algorithm C, where the time complexity is $O(n^2 \log n)$ - the fastest algorithm when n increases.

3. Comparison Table

Size (n)	Quadratic (ms)	Subquadratic (ms)
2	0.0122	0.0096
4	0.0235	0.0390
8	0.0768	0.1281
16	0.3104	0.3931
32	1.2492	1.5047
64	5.0236	5.7959
128	19.9760	22.9653
256	80.6061	91.3788